

■ Sales Forecasting and Demand Prediction Using Machine Learning

Executive Summary

Accurate sales forecasting plays a crucial role in business decision-making, especially in inventory planning, supply chain management, and revenue optimization. This project focuses on developing a machine learning–based time series forecasting model to predict future sales using historical retail data. The dataset used in this project contains order-level sales information including dates, customer details, product categories, and sales values.

The raw data was cleaned and transformed into monthly aggregated sales data to capture long-term trends and seasonal patterns. A time series forecasting approach was applied using the Holt-Winters Exponential Smoothing model, which is suitable for data exhibiting both trend and seasonality. Model performance was evaluated using standard forecast accuracy metrics such as MAE, RMSE, and MAPE.

The final model achieved a Mean Absolute Percentage Error (MAPE) of approximately **14.48%**, which indicates good forecasting accuracy for retail sales data. The forecast results were visualized using Power BI dashboards to support business interpretation. Based on the forecast trends, practical recommendations were provided to assist management in inventory control and demand planning.

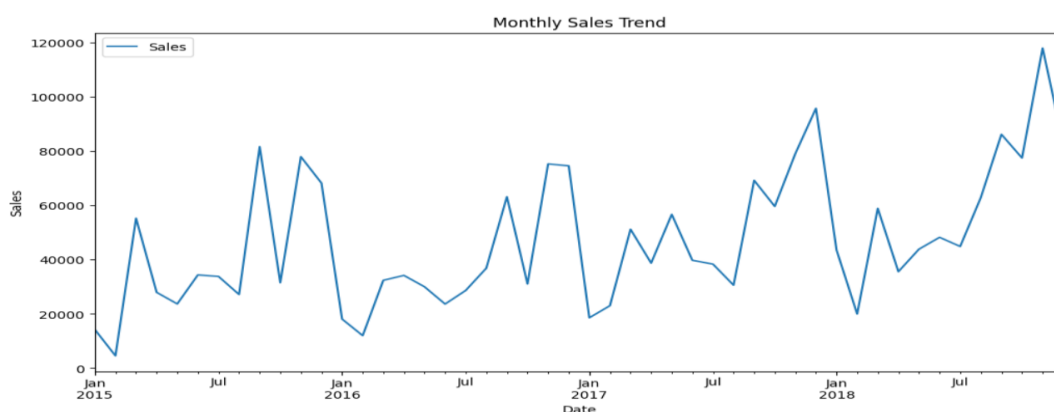
Machine Learning Model Explanation

Data Preparation

The original dataset contained daily sales transactions. To make the data suitable for time series forecasting, the following preprocessing steps were performed:

- Removal of unnecessary columns
- Conversion of date columns into proper datetime format
- Aggregation of sales values at a **monthly level**
- Sorting data chronologically

Monthly aggregation helps reduce noise and highlights seasonal sales patterns more clearly.



Model Selection

The **Holt-Winters Exponential Smoothing** model was chosen for this project because:

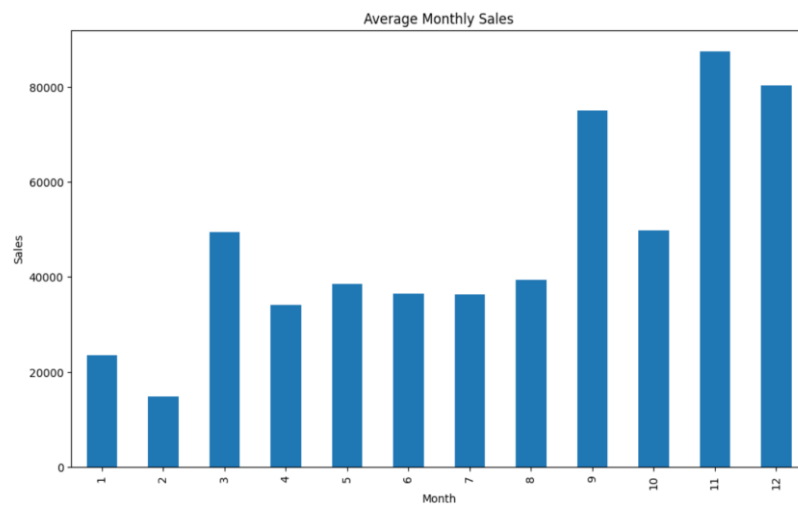
- It effectively models **trend** and **seasonality**

- It performs well on retail sales data
- It is interpretable and widely used in industry

The model components include:

- **Trend component:** captures long-term increase or decrease in sales
- **Seasonal component:** captures recurring monthly patterns
- **Error component:** handles random fluctuations

A multiplicative seasonal approach was considered suitable as seasonal effects increase with higher sales values.

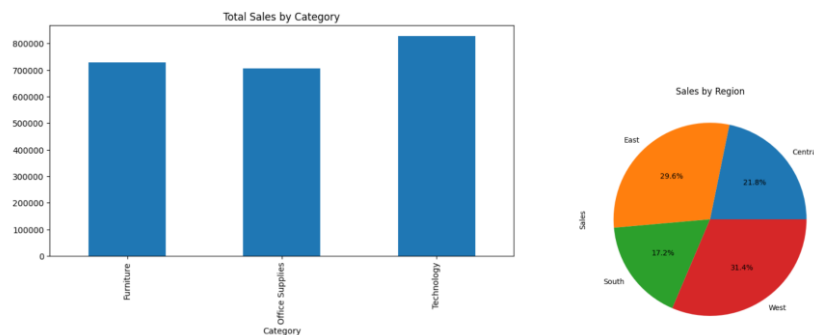


Training and Forecasting

The dataset was divided into:

- **Training set:** historical sales data
- **Test set:** last 6 months of data for validation

The model was trained on the training set and used to forecast future monthly sales. The predicted values were compared with actual sales to evaluate accuracy.



Forecast Accuracy Metrics

To evaluate the forecasting performance, three standard metrics were used:

1. Mean Absolute Error (MAE)

MAE measures the average absolute difference between actual and predicted sales values.

- **MAE $\approx 5,770$**

This indicates that the model's predictions differ from actual sales by approximately 5,770 units on average.

2. Root Mean Squared Error (RMSE)

RMSE penalizes larger errors more heavily and highlights months with higher deviation.

- **RMSE $\approx 7,272$**

This value suggests the presence of some high-variance months, likely due to seasonal peaks or promotional periods.

3. Mean Absolute Percentage Error (MAPE)

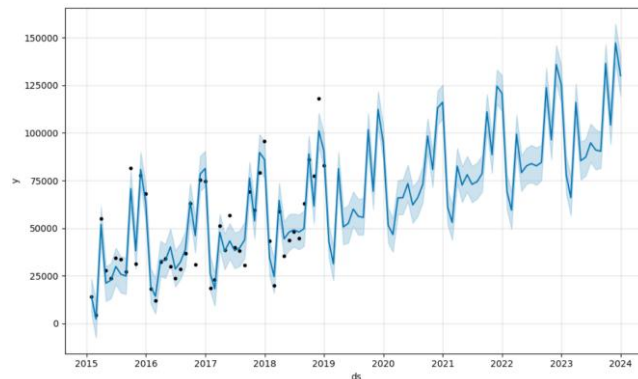
MAPE expresses forecast error as a percentage, making it easy to interpret.

- **MAPE $\approx 14.48\%$**

According to industry standards:

- MAPE below 20% is considered **good**
- Below 10% is considered excellent

Therefore, the developed model provides **reliable and business-acceptable forecasts**.

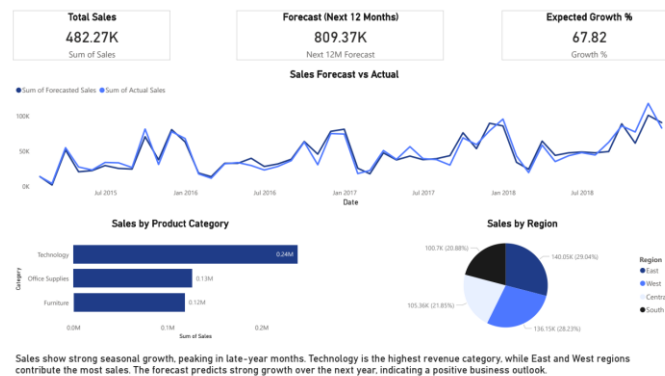


Dashboard Visualization (Power BI)

Power BI was used to visualize and interpret the forecasted results. The dashboard includes:

- Line chart showing **Actual vs Forecasted Sales**
- Monthly sales trend analysis
- Forecast horizon highlighting future demand
- Clear comparison between historical and predicted values

These visualizations make the results easily understandable for non-technical stakeholders and support management-level decision-making.



Recommendations for Management

Based on the analysis and forecast results, the following recommendations are proposed:

- 1. Inventory Planning**
Increase inventory levels before forecasted peak months to avoid stockouts.
- 2. Demand-Based Procurement**
Use forecasted sales values to plan supplier orders and reduce excess inventory costs.
- 3. Seasonal Strategy**
Identify high-demand months and align promotions, discounts, and marketing campaigns accordingly.
- 4. Data-Driven Decision Making**
Regularly update the forecasting model with new sales data to improve accuracy over time.
- 5. Category-Level Forecasting (Future Scope)**
Future improvements may include forecasting at category or product level for finer-grained planning.

Conclusion

This project successfully demonstrates the application of machine learning techniques for sales forecasting and demand prediction. By using historical sales data and Holt-Winters time series modeling, meaningful insights into future demand were obtained. The model achieved good accuracy, as reflected by a MAPE of approximately 14.48%, making it suitable for real-world business use.

The integration of Power BI dashboards further enhances the practical value of the project by enabling intuitive data visualization and decision support. Overall, this project highlights how data science and machine learning can help organizations improve operational efficiency and strategic planning.